

Ques 2-1 Soph Set - A

Ans 1: Karl Pearson's Coefficient of Skewness = 0.64
 $Sk = 0.64$

Standard Deviation = $\sigma = 13$

mean = 59.2

$$Sk = \frac{\text{mean} - \text{mode}}{\sigma} \Rightarrow 0.64 = \frac{59.2 - \text{mode}}{13}$$

$$\Rightarrow \text{mode} = 50.88$$

$$\text{mode} = 3 \text{ median} - 2 \text{ mode}$$

$$50.88 = 3 \text{ median} - 2 \times 59.2 \Rightarrow \text{median} = 56.42$$

$$\therefore (\text{mode}, \text{median}) = (50.88, 56.42) \quad \boxed{A}$$

Ans 2: Standard Deviation = $\sqrt{u_2} = 10.95$

$$\Rightarrow u_2 = (10.95)^2 = 120$$

for distn to be platykurtic $\beta_2 = \frac{u_4}{u_2^2} < 3$

$$\Rightarrow u_4 < 3 u_2^2$$

$$\Rightarrow u_4 < 3 \times (120)^2 \Rightarrow \boxed{u_4 < 43,200} \quad \boxed{C}$$

Ans 3: Let E_1 : Event that a randomly selected bolt is manufactured by machine A

E_2 : Event that a randomly selected bolt is manufactured by machine B

E : Event that chosen bolt is defective

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So $P(E_1) = 0.25$, $P(E_2) = 0.35$, $P(E_3) = 0.40$

$P(E|E_1) = 0.05$, $P(E|E_2) = 0.04$, $P(E|E_3) = 0.02$

The probability that a defective bolt selected at random is manufactured by machine A = $P(E_1|E)$

$$P(E_1|E) = \frac{P(E_1)P(E|E_1)}{\sum_{i=1}^3 P(E_i)P(E|E_i)}$$

$$= \frac{0.25 \times 0.05}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02} = \frac{125}{345}$$

$$= \frac{25}{69} \quad \downarrow \text{Let } T$$

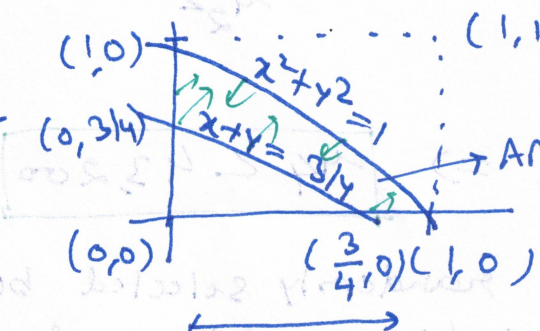
$$P(E_2|E) = \frac{0.35 \times 0.04}{T} = \frac{140}{345} = \frac{28}{69}$$

~~$P(E_3|E) = \frac{0.40 \times 0.02}{T}$~~

$$P(E_3|E) = 1 - [P(E_1|E) + P(E_2|E)] = 1 - \frac{25}{69} - \frac{28}{69} = \frac{16}{69}$$

$\left(\frac{25}{69}, \frac{28}{69}, \frac{16}{69} \right)$ D

Ans 4:



$$P(A|B) = \frac{\text{Area}(A \cap B)}{\text{Area}(B)}$$

$$= \frac{\frac{\pi}{4} - \frac{1}{2} \times \frac{3}{4} \times \frac{3}{4}}{\frac{1}{4} \times \pi + 1}$$

$$= 0.6419 \quad \text{[A]}$$

Ans 5: let P = Probability (no car in 10 min)

$P^2 = P \times P = \text{Pro (no car in 10 min)} \times \text{Pro (no car in 10 min)}$

$\therefore \text{Pro (no car in 30 min)} = P^3 \Rightarrow \text{Pro (car in 30 min)} = 1 - P^3 = 0.95$

$\Rightarrow P = 0.368$

$\Rightarrow \text{Pro (no car in 10 min)} = 0.368$

$\Rightarrow \text{Pro (car in 10 min)} = 1 - 0.368 = 0.632$ Any.

0.632, D

$$\begin{aligned}
 6. \quad P(0.2 \leq x \leq 0.3) &= \int_{0.2}^{0.3} (12.5x - 1.25) dx \\
 &= \left[12.5 \frac{x^2}{2} - 1.25x \right]_{0.2}^{0.3} \\
 &= \frac{12.5}{2} (0.3^2 - 0.2^2) - 1.25 \times 0.1 \\
 &= \boxed{0.1875}
 \end{aligned}$$

7. The probability mass function of the RV X is determined as follows

$$P(X=1) = \frac{1}{2} - 0 = \frac{1}{2}$$

$$P(X=2) = \frac{5}{6} - \frac{1}{2} = \frac{2}{6}$$

$$P(X=3) = 1 - \frac{5}{6} = \frac{1}{6}$$

$$\therefore E(X) = \sum x P(X=x)$$

$$= 1 \cdot \frac{1}{2} + 2 \cdot \frac{2}{6} + 3 \cdot \frac{1}{6}$$

$$\boxed{E(X) = \frac{5}{3}}$$

$$\begin{aligned}
 8. \quad M(t) &= \frac{e^t - 1}{t} \\
 &= \frac{1}{t} \left[1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots - 1 \right] \\
 &= 1 + \frac{t}{2!} + \frac{t^2}{3!} + \frac{t^3}{4!} + \dots
 \end{aligned}$$

$$\therefore E(X) = \frac{1}{2}, \quad E(X^2) = \frac{1}{3} \quad \leftarrow (\text{coefficient of } \frac{t^2}{2!})$$

$$\begin{aligned}
 \therefore V(X) &= E(X^2) - (E(X))^2 \\
 &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12}
 \end{aligned}$$

$$\therefore \boxed{V(X) = \frac{1}{12}}$$

9. The marginal density of x is

$$\begin{aligned} f_x(x) &= \int_{-\infty}^{\infty} f(x,y) dy \\ &= \int_0^2 \left(x^2 + \frac{xy}{3} \right) dy \\ &= \left[x^2 y + \frac{xy^2}{6} \right]_{y=0}^2 \end{aligned}$$

$$\sigma, \quad \boxed{f_x(x) = 2x^2 + \frac{2x}{3}, \quad 0 \leq x \leq 1}$$

10.

$$E(I) = E(I_0(e^{5V} - 1))$$

$$= I_0 \{ E(e^{5V}) - 1 \}$$

$$= I_0 \left\{ \int_1^3 e^{5v} \cdot \frac{1}{2} dv - 1 \right\}$$

$$= I_0 \left\{ \left[\frac{e^{5v}}{10} \right]_1^3 - 1 \right\}$$

$$\sigma, \quad \boxed{E(I) = \frac{I_0}{10} \{ e^{15} - e^5 - 10 \}}$$